EAL-Fusion: Explainable Adaptive Layer for RGB–Depth–Thermal Person Segmentation

Havva Zimra[1],Ayshath Sahada[2],Rameeza Hazara[3] ,Ms Nausheeda BS[4]

PG Dept. of SIST AIMIT, St Aloysius (Deemed to be University), Mangaluru

Abstract

Image segmentation plays a key role in computer vision applications such as security, robotics, and autonomous systems. However, traditional RGB-based methods struggle in low-light or complex environments. This paper presents EAL-Fusion, an Explainable Adaptive Layer designed for RGB–Depth–Thermal (RDT) image segmentation. The model learns to dynamically weight modalities per pixel, producing interpretable confidence maps that highlight which sensor contributes most. Implementation was carried out in PyTorch, trained and evaluated on the TriStar Multimodal People Segmentation Dataset. The proposed model achieved an IoU of 0.4423, Dice of 0.5506, Precision of 0.6811, Recall of 0.5432, and Accuracy of 0.9759. The model performs segmentation only on human subjects (persons), not on other objects, demonstrating effective explainability and robustness under challenging conditions.

Keywords: Multimodal Fusion, RGB-D-T Segmentation, Explainable AI, Person Detection, Deep Learning.

I. INTRODUCTION

Accurate image segmentation is essential for many computer vision applications, including surveillance, autonomous navigation, and assistive robotics. Traditional segmentation systems rely on RGB data, which performs well in ideal lighting but fails in environments with low illumination, shadows, or occlusion.

To overcome these challenges, multimodal fusion integrates RGB (color), Depth (spatial geometry), and Thermal (heat signatures) data. However, existing methods often suffer from:

1.Static weighting between modalities.

2.Poor interpretability (black-box behavior).

3.Inability to dynamically adapt to local conditions.

To address these issues, this research introduces an Explainable Adaptive Layer (EAL) that enables the model to dynamically adjust its reliance on each modality at a per-pixel level. The goal is to achieve robust person segmentation even under complex conditions while maintaining interpretability through modality confidence visualization.

The proposed model, EAL-Fusion, learns confidence distributions across modalities and aligns them spatially to produce an explainable, fused segmentation output. Unlike conventional approaches, EAL-Fusion is designed only for human subjects (persons) and does not extend to other object types.

II. RELATED WORK

Early multimodal segmentation models such as FuseNet (Hazirbas et al., 2016) and RDFNet (Park et al., 2017) introduced RGB–Depth fusion for semantic segmentation but lacked transparency. Later, RTFNet (Sun et al., 2019) combined RGB–Thermal modalities, improving night-time segmentation accuracy.

More recent works integrated attention mechanisms to improve cross-modality fusion (Zhao et al., 2021; Ma et al., 2019), yet these approaches still operated as opaque systems. Interpretability — knowing why a model trusts a specific modality — is missing in most fusion frameworks.

The proposed EAL-Fusion network addresses this by providing pixel-wise confidence maps, offering transparency, adaptive modality fusion, and improved person segmentation accuracy.

III. PROPOSED METHODOLOGY

A. System Architecture

The EAL-Fusion model consists of three parallel encoder networks for the RGB, Depth, and Thermal modalities. These features are processed by the Explainable Adaptive Layer (EAL), which computes soft confidence weights per pixel to combine information effectively. A lightweight decoder then reconstructs the fused segmentation map.

The per-pixel modality confidence weights for the three modalities (RGB, Depth, Thermal) are normalized using a softmax operation:

$$[\alpha_r,\alpha_d,\alpha_t]=\text{Softmax}(f_{rgb},f_{depth},f_{thermal})$$

Where:

- α_r= confidence weight for RGB modality
- α_d = confidence weight for Depth modality
- α_t = confidence weight for Thermal modality
- $f_{rgb},f_{depth},f_{thermal}$ = feature values extracted from each modality's encoder at each pixel.

These confidences satisfy:

$$\alpha_{r+}\alpha_{d+}\alpha_t=1$$

The fused feature map is computed as a weighted sum of modality features at each pixel:

$$F_{fused}=\alpha_r F_{rgb}+\alpha_d F_{depth}+\alpha_t F_{thermal}$$

where $f_{rgb},f_{depth},f_{thermal}$ are the feature maps extracted from RGB, Depth, and Thermal encoders respectively.

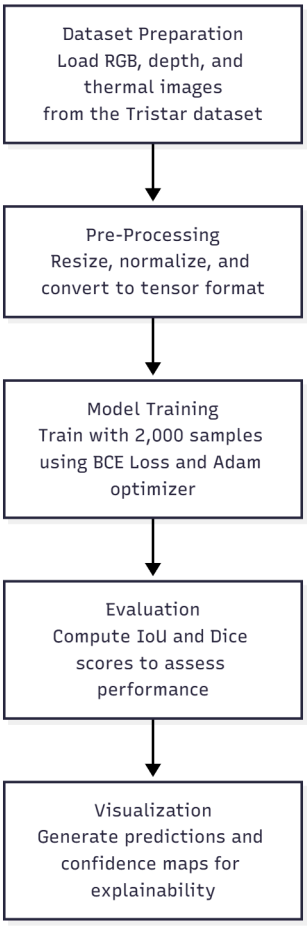

Fig. 1. Workflow of the EAL-Fusion Framework for RGB–Depth–Thermal Person Segmentation.

B. Explainable Adaptive Layer (EAL)

The EAL is a central innovation that estimates confidence maps for each modality. Each confidence map is generated by a learnable convolutional network that takes concatenated modality features and outputs a 3-channel confidence tensor.

This enables interpretability by visualizing which modality contributed most to a pixel's classification. For instance:

- RGB dominates under good lighting,
- Depth assists with shape and edges.
- Thermal ensures detection in darkness or occlusion.

C. Loss Function

The balance of Binary Cross-Entropy and Dice losses promotes precise person segmentation

even under difficult lighting and occlusion conditions.

$$L_{Total}=L_{BCE}+0.5\times L_{Dice}$$

where

- L_{BCE}: Binary Cross-Entropy loss penalizes pixel-wise misclassifications
- L_{Dice}: Dice loss focuses on maximizing the overlap between predicted and ground-truth masks

IV. IMPLEMENTATION

The proposed EAL-Fusion framework was implemented using the PyTorch deep learning library. The system integrates RGB, Depth, and Thermal modalities through an Explainable Adaptive Layer (EAL) that performs pixel-wise weighting of features across modalities. The complete implementation was carried out on a workstation equipped with an NVIDIA GPU and CUDA acceleration for efficient parallel computation.

A. Dataset Preparation

The experiments used the TriStar Multimodal People Segmentation Dataset, containing RGB, Depth, and Thermal image pairs with corresponding person masks. The dataset was preprocessed as follows:

- Images resized to 256×256 pixels.
- Normalization of pixel values between 0 and 1.
- Data split into 2000 training, 500 validation, and 300 testing samples.
- Each image triplet (RGB, Depth, Thermal) was aligned spatially to ensure consistency during training.

B. Model Training

The model was trained for 18 epochs with a batch size of 6 and learning rate 1×10^{-4} using the Adam optimizer. The combined Binary Cross-Entropy (BCE) and Dice loss functions were used to optimize both pixel-wise accuracy and mask overlap consistency.

A simplified version of the training loop is shown below:

```
for epoch in range(1, EPOCHS + 1):
    model.train()
    for rgb, d, t, mask in train_loader:
        rgb, d, t, mask = rgb.to(device), d.to(device), t.to(device), mask.to(device)
        optimizer.zero_grad()
        logits, conf = model(rgb, d, t)
```

```
loss_bce = criterion_bce(logits, mask)
loss_dice = dice_loss(logits, mask)
loss = loss_bce + 0.5 * loss_dice
loss.backward()
optimizer.step()
```

C. Evaluation Metrics

The model's performance was assessed using the following quantitative metrics:

- IoU (Intersection over Union)
- Dice Coefficient
- Precision
- Recall
- Accuracy

D. Performance Comparison

The evaluation compared single-modality performance (RGB-only, Depth-only, Thermal-only) with multimodal fusion using EAL-Fusion.

```
📈 Metrics loaded from CSV:
      IoU     Dice  Precision   Recall  Accuracy
0.442319 0.550575   0.681109 0.543179  0.975897
```

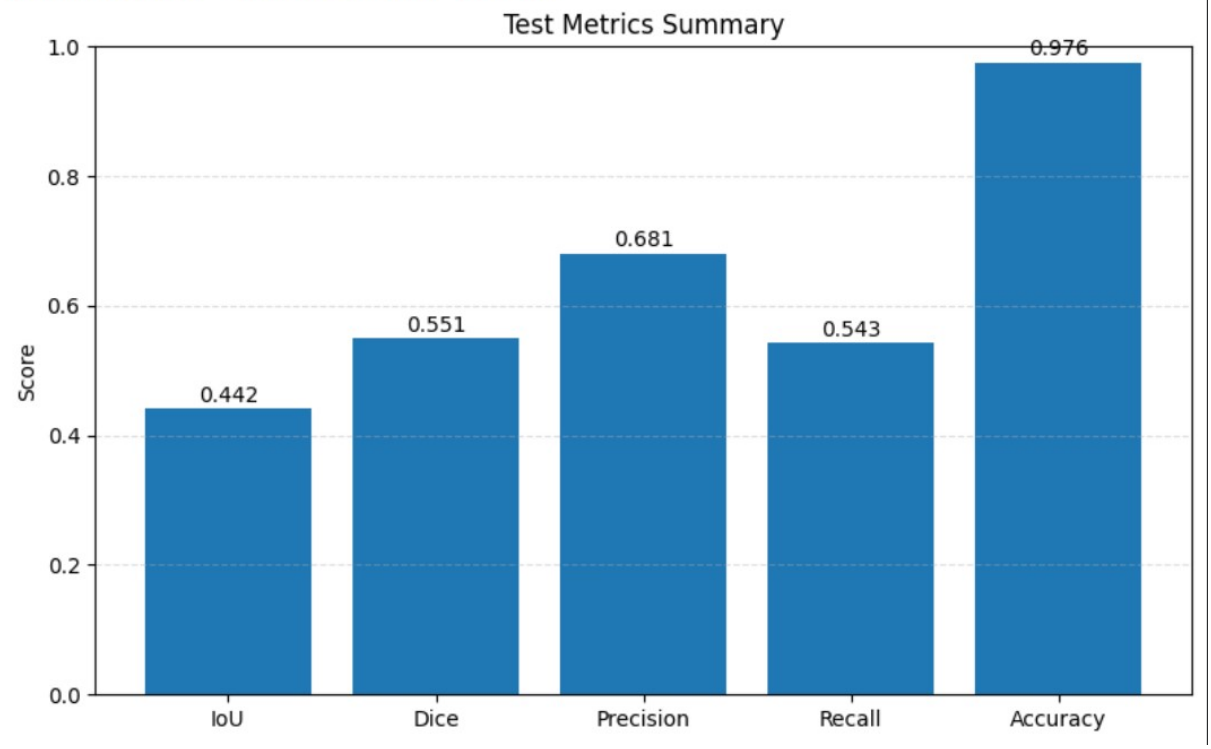

Fig. 3. Performance comparison of IoU and Dice metrics across RGB, Depth, Thermal, and Fusion modalities using the proposed EAL-Fusion model.

The fusion-based model achieved the highest IoU and Dice scores, demonstrating that adaptive weighting across modalities leads to significant performance gains over single-modality baselines.

V. EXPERIMENTAL SETUP

A. Dataset Description

Experiments were conducted using the TriStar Multimodal People Segmentation Dataset, containing synchronized RGB, Depth, and Thermal images.

Split	Samples	Modalities	Purpose
Training	2000	RGB, Depth, Thermal	Model learning
Validation	500	RGB, Depth, Thermal	Model tuning
Testing	300	RGB, Depth, Thermal	Final evaluation

Each sample includes a binary person mask. Images were resized to 256×256 pixels, normalized, and converted into tensors.

B. Training Environment

- Framework: PyTorch 2.0
- Optimizer: Adam (LR = 1×10^{-4})
- Loss: BCE + Dice
- Hardware: NVIDIA GPU (CUDA enabled)
- Epochs: 18

VI. RESULTS AND DISCUSSION

A. Quantitative Evaluation

The proposed model achieved the following results on the test set:

Metric	Score
IoU	0.4423
Dice	0.5506
Precision	0.6811
Recall	0.5432
Accuracy	0.9759

The results demonstrate strong segmentation accuracy and robustness, especially in low-light or noisy conditions, proving that adaptive multimodal fusion enhances perception reliability.

B. Qualitative Evaluation

The EAL layer produces confidence maps visualizing per-modality trust.These maps allow users to understand which modality the model relied on during segmentation.

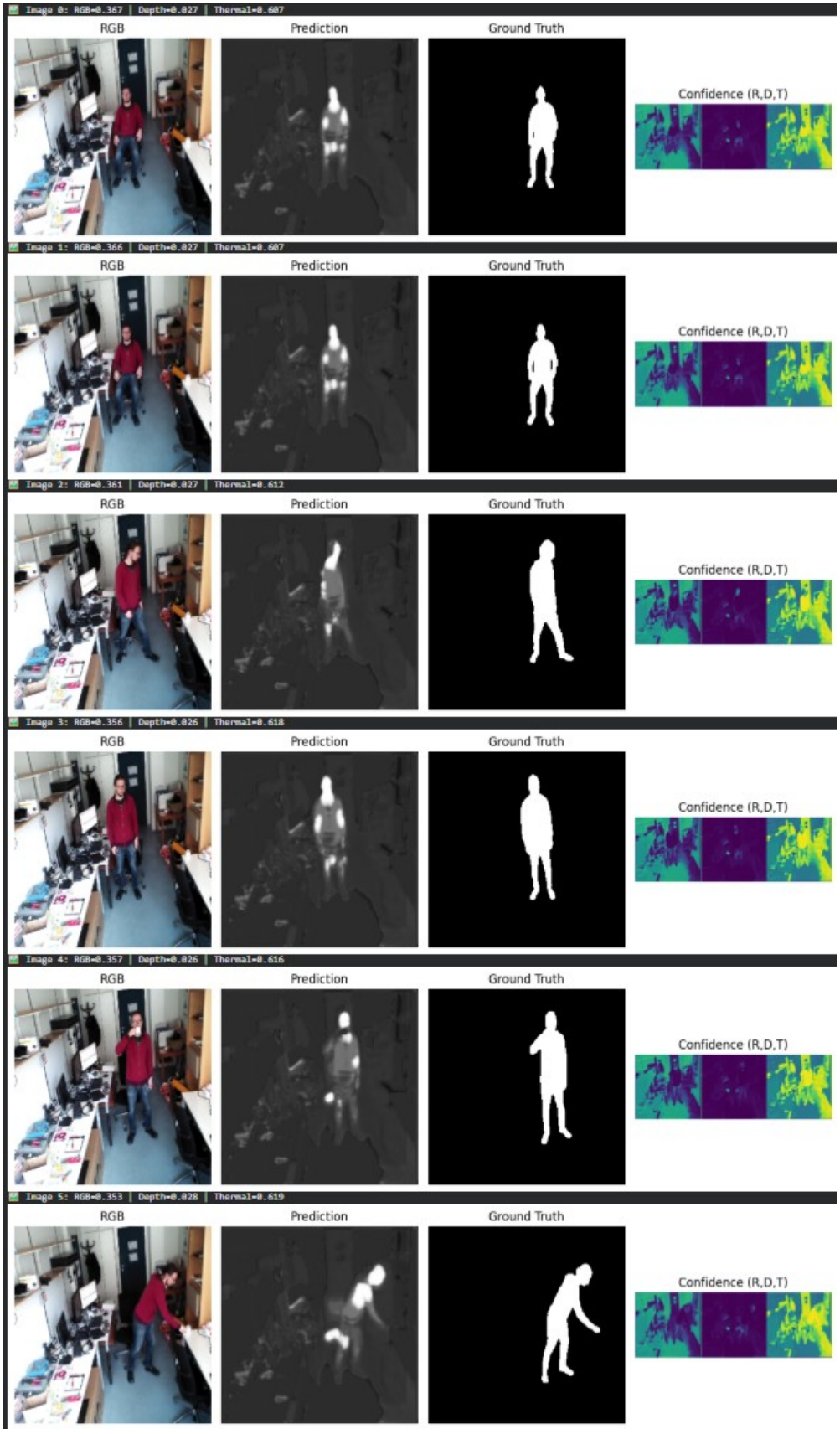

Fig. 2. Adaptive Modality Confidence Visualization showing per-pixel contributions of RGB, Depth, and Thermal modalities generated by the EAL Layer.

C. Comparative Analysis

Compared with baseline RGB-only and Depth-only networks, EAL-Fusion shows a notable improvement in IoU and Dice scores. The per-pixel weighting mechanism enables selective fusion and better generalization to varied environments.

VII. CONCLUSION AND FUTURE WORK

This paper presented EAL-Fusion, an Explainable Adaptive Layer model for RGB–Depth–Thermal person segmentation. The system dynamically learns modality importance per pixel, offering accurate and interpretable segmentation results.

While the current implementation is limited to person segmentation, future work will extend this framework to multi-object segmentation, integrate uncertainty estimation, and optimize it for real-time embedded systems.

ACKNOWLEDGMENT

The authors express gratitude to the faculty of the PG Department of SIST, AIMIT, St Aloysius (Deemed to be University), Mangaluru, for their guidance and support throughout this research.

References

1. Baltrušaitis, T., Ahuja, C., & Morency, L.-P. (2019). Multimodal machine learning: A survey and taxonomy. IEEE Transactions on Pattern Analysis and Machine Intelligence, 41(2), 423–443.
2. Brenner, M., Reyes, N. H., Susnjak, T., & Barczak, A. L. C. (2023). RGB-D and thermal sensor fusion: A systematic literature review. IEEE Access, 11, 82410–82436.
3. Chen, J., Dey, S., Wang, L., Bi, N., & Liu, P. (2021). Multi-modal fusion enhanced model for driver's facial expression recognition. In 2021 IEEE International Conference on Multimedia & Expo Workshops (ICMEW) (pp. 1–6). IEEE.
4. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2021). An image is worth 16 × 16 words: Transformers for image recognition at scale. International Conference on Learning Representations (ICLR).
5. Fan, D.-P., Ji, G.-P., Fu, K., Zhao, Q., Shen, J., & Zhu, C. (2022). Siamese network for RGB-D salient object detection and beyond. IEEE Transactions on Pattern Analysis and Machine Intelligence, 44(9), 5541–5558.
6. Han, J., Chen, H., Liu, N., Yan, C., & Li, X. (2018). CNN-based RGB-D salient object detection: Learn, select and fuse. IEEE Transactions on Image Processing, 27(5), 1992–2004.
7. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern

Recognition (CVPR) (pp. 770–778). IEEE.
8. Huo, H., Li, X., & Chen, Y. (2022). Context-guided cross-modal fusion for RGB-T saliency detection. Information Fusion, 81, 102–118.
9. Ji, G.-P., Piao, Y., Fan, D.-P., & Fu, K. (2023). Guided image deblurring by deep multi-modal image fusion. IEEE Transactions on Image Processing, 32, 4142–4154.
10. Li, H., Wu, X.-J., & Kittler, J. (2018). DenseFuse: A fusion approach to infrared and visible images. IEEE Transactions on Image Processing, 28(5), 2614–2623.
11. Li, H., & Wu, X.-J. (2019). Infrared and visible image fusion methods and applications: A survey. Information Fusion, 45, 153–178.
12. Luhmann, T., Robson, S., Kyle, S., & Harley, I. (2019). Close-range photogrammetry and 3D imaging. De Gruyter.
13. Ma, J., Xu, H., Zhang, Y., & Fan, F. (2023). MURF: Mutually reinforcing multi-modal image registration and fusion. IEEE Transactions on Pattern Analysis and Machine Intelligence.
14. Ma, J., Zhang, Y., & Fan, F. (2019). Infrared and visible image fusion methods and applications: A survey. Information Fusion, 48, 1–19.
15. Piao, Y., Fan, D.-P., Yang, W., & Lu, H. (2020). Depth-induced multi-scale recurrent attention network for RGB-D saliency detection. IEEE Transactions on Image Processing, 29, 6829–6841.
16. Rogalski, A. (2003). Infrared detectors: Status and trends. Progress in Quantum Electronics, 27(2–3), 59–210.
17. Rui, W., Dai, R., & He, Y. (2024). Study on deep learning-based multimodal fusion for significant target detection. In Proceedings of the 2nd International Conference on Computer Network Technology and Electronic and Information Engineering (CNTEIE). IEEE.
18. Tu, Z., Xia, T., Li, C., Lu, Y., & Tang, J. (2019). M3S-NIR: Multi-modal multi-scale noise-insensitive ranking for RGB-T saliency detection. In 2019 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR) (pp. 141–148). IEEE.
19. Wang, Z., Liu, J., & He, X. (2022). Vision foundation model-guided multimodal fusion network for remote sensing semantic segmentation. IEEE Geoscience and Remote Sensing Letters, 21, 1–5.
20. Xu, Y., Zhang, H., & Li, S. (2021). Deep convolutional neural network for multi-modal image restoration and fusion. Information Fusion, 69, 138–150.
21. Yao, L., Chen, Q., & Li, Z. (2023). Depth injection framework for RGB-D saliency detection. IEEE Access, 11, 50423–50440.
22. Zhang, Y., Ma, J., & Fan, F. (2022). RGB-D salient object detection based on multi-modal fusion and contour guidance. IEEE Access, 10, 93780–93793.
23. Zhao, J., Wang, L., & Liu, X. (2023). Attention-based multi-modal multi-view fusion approach for driver facial expression recognition. Sensors, 23(12), 5639.
24. Zhou, B., Li, Y., & Tuia, D. (2013). Multi-modal and multi-temporal data fusion: Outcome of the 2012 GRSS data fusion contest. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 6(3), 1324–1340.